

$$K = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) \quad V = mgz$$

3a. $\mathcal{L}(x, y, z, \dot{x}, \dot{y}, \dot{z}) = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} \quad \frac{\partial \mathcal{L}}{\partial x} = 0 \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) - \frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\frac{d}{dt}(m\dot{x}) = 0$$

$$\frac{\partial \mathcal{L}}{\partial \dot{y}} = m\dot{y} \quad \frac{\partial \mathcal{L}}{\partial y} = 0 \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{y}}\right) - \frac{\partial \mathcal{L}}{\partial y} = 0 \quad \frac{d}{dt}(m\dot{y}) = 0$$

$$\frac{\partial \mathcal{L}}{\partial \dot{z}} = m\dot{z} \quad \frac{\partial \mathcal{L}}{\partial z} = -mg \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{z}}\right) - \frac{\partial \mathcal{L}}{\partial z} = 0$$

$$\frac{d}{dt}(m\dot{z}) + mg = 0 \quad \frac{d}{dt}(\dot{z}) = -g$$

$$\boxed{\ddot{x} = 0 \quad \ddot{y} = 0 \quad \ddot{z} = -g}$$

3b. $K = \frac{1}{2}m\dot{x}^2 \quad V = \frac{1}{2}kx^2 \quad \mathcal{L}(x, \dot{x}) = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} \quad \frac{\partial \mathcal{L}}{\partial x} = -\frac{1}{2}k(2x) = -kx \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) - \frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\frac{d}{dt}(m\dot{x}) + kx = 0 \quad m\ddot{x} + kx = 0 \quad \boxed{\ddot{x} = -\frac{k}{m}x}$$

3c. $K = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \quad V = \frac{1}{2}k(x^2 + y^2)$

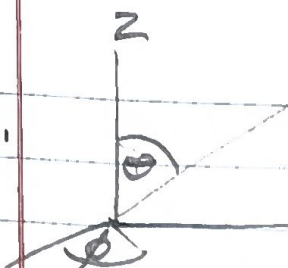
$$\mathcal{L}(x, y, \dot{x}, \dot{y}) = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - \frac{1}{2}k(x^2 + y^2)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} \quad \frac{\partial \mathcal{L}}{\partial x} = -\frac{1}{2}k(2x) = -kx \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) - \frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\frac{d}{dt}(m\dot{x}) + kx = 0 \quad \boxed{\ddot{x} = -\frac{k}{m}x}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{y}} = m\dot{y} \quad \frac{\partial \mathcal{L}}{\partial y} = -\frac{1}{2}k(2y) = -ky \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{y}}\right) - \frac{\partial \mathcal{L}}{\partial y} = 0$$

$$\frac{d}{dt}(m\dot{y}) + ky = 0 \quad \boxed{\ddot{y} = -\frac{k}{m}y}$$

3d.  (r, θ, ϕ) $x = r \sin \theta \cos \phi$ $y = r \sin \theta \sin \phi$ $z = r \cos \theta$
The position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z} = r\hat{r}$

$\vec{r} = r(\sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z})$

Then $\dot{\vec{r}} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\hat{r}) = r\frac{d\hat{r}}{dt} + \frac{dr}{dt}\hat{r}$

$\hat{r} = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}$

The unit vector $\hat{r}$ is a function of θ and ϕ where θ and ϕ are functions of time t , $\theta(t)$, $\phi(t)$. Then we can use multivariable chain rule for $\frac{d\hat{r}}{dt}$

$\frac{d\hat{r}}{dt} = \frac{\partial \hat{r}}{\partial \theta} \frac{d\theta}{dt} + \frac{\partial \hat{r}}{\partial \phi} \frac{d\phi}{dt}$

$\frac{\partial \hat{r}}{\partial \theta} = \cos \theta \cos \phi \hat{x} + \cos \theta \sin \phi \hat{y} - \sin \theta \hat{z}$

$\frac{\partial \hat{r}}{\partial \theta}$ is the tangent vector pointing in the direction of increasing θ $\frac{\partial \hat{r}}{\partial \theta} = \left| \frac{\partial \hat{r}}{\partial \theta} \right| \hat{\theta}$

$\left| \frac{\partial \hat{r}}{\partial \theta} \right| = \sqrt{\cos^2 \theta \cos^2 \phi + \cos^2 \theta \sin^2 \phi + \sin^2 \theta}$

$= \sqrt{\cos^2 \theta (\cos^2 \phi + \sin^2 \phi) + \sin^2 \theta} = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$

The magnitude is 1 so $\frac{\partial \hat{r}}{\partial \theta}$ is the unit vector $\hat{\theta}$

$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}$

$\frac{\partial \hat{r}}{\partial \phi} = -\sin \theta \sin \phi \hat{x} + \sin \theta \cos \phi \hat{y}$

$= \sin \theta (-\sin \phi \hat{x} + \cos \phi \hat{y}) = \sin \theta \hat{\phi}$

Since $\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$ from polar coordinates

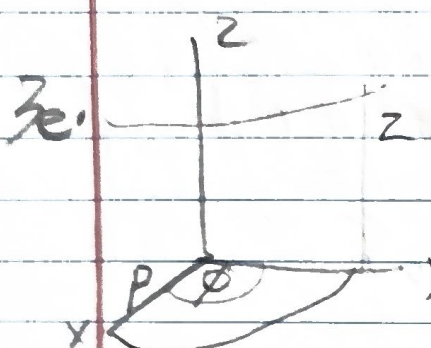
$\frac{d\hat{r}}{dt} = \dot{\theta} \hat{\theta} + \dot{\phi} \sin \theta \hat{\phi}$

$$\vec{v} = \frac{d\vec{r}}{dt} = r(\dot{\theta}\hat{\theta} + \dot{\phi}\sin\theta\hat{\phi}) + \dot{r}\hat{r}$$

$$= \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\dot{\phi}\sin\theta\hat{\phi}$$

$$|\vec{v}| = \sqrt{\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2}$$

$$T = \frac{1}{2}m|\vec{v}|^2 = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2)$$


 The position vector $\vec{r}$
 $x = r\cos\phi$
 $y = r\sin\phi$
 $z = z$
 $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$
 $= r\cos\phi\hat{x} + r\sin\phi\hat{y} + z\hat{z}$
 $\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\cos\phi\hat{x} + r\sin\phi\hat{y} + z\hat{z})$

The unit vector $\hat{x}, \hat{y}, \hat{z}$ are constants with time t since they never change direction.

$$\vec{v} = (\dot{r}\cos\phi - r\sin\phi\dot{\phi})\hat{x} + (\dot{r}\sin\phi + r\cos\phi\dot{\phi})\hat{y} + \dot{z}\hat{z}$$

$$= \dot{r}\cos\phi\hat{x} - r\sin\phi\dot{\phi}\hat{x} + \dot{r}\sin\phi\hat{y} + r\cos\phi\dot{\phi}\hat{y} + \dot{z}\hat{z}$$

$$= \dot{r}(\cos\phi\hat{x} + \sin\phi\hat{y}) + r\dot{\phi}(\cos\phi\hat{y} - \sin\phi\hat{x}) + \dot{z}\hat{z}$$

But $\hat{p} = \cos\phi\hat{x} + \sin\phi\hat{y}$, $\hat{\phi} = -\sin\phi\hat{x} + \cos\phi\hat{y}$
 from polar coord

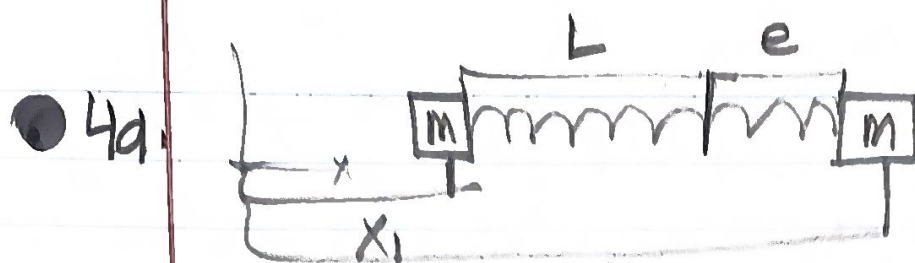
$$= \dot{r}\hat{p} + r\dot{\phi}\hat{\phi} + \dot{z}\hat{z}$$

$$\vec{v} = \dot{r}\hat{p} + r\dot{\phi}\hat{\phi} + \dot{z}\hat{z} \quad |\vec{v}| = \sqrt{\dot{r}^2 + r^2\dot{\phi}^2 + \dot{z}^2}$$

$$T = \frac{1}{2}m|\vec{v}|^2 = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2 + \dot{z}^2)$$

$$\frac{d\sin\phi}{dt} = \cos\phi\dot{\phi}$$

$$\frac{d\cos\phi}{dt} = -\sin\phi\dot{\phi}$$



$$x_1 + L + e = x_2$$

$$e = x_2 - x_1 - L$$

$$U = \frac{1}{2} k e^2 = \frac{1}{2} k (x_2 - x_1 - L)^2 = \frac{1}{2} k (x_1 + L - x_2)^2$$

$$T = \frac{1}{2} m (\dot{x}_1^2 + \dot{x}_2^2)$$

4b. $\mathcal{L}(x_1, x_2, \dot{x}_1, \dot{x}_2) = T - U = \frac{1}{2} m (\dot{x}_1^2 + \dot{x}_2^2) - \frac{1}{2} k (x_1 + L - x_2)^2$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}_1} = m \dot{x}_1, \quad \frac{\partial \mathcal{L}}{\partial x_1} = -\frac{1}{2} k \frac{d(x_1 + L - x_2)^2}{d(x_1 + L - x_2)} \frac{\partial (x_1 + L - x_2)}{\partial x_1} = -k(x_1 + L - x_2)$$

$$\frac{d}{dt}(m \dot{x}_1) + k(x_1 + L - x_2) = 0$$

$$\ddot{x}_1 = -\frac{k}{m}(x_1 + L - x_2) = \frac{k}{m}(x_2 - x_1 - L)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}_2} = m \dot{x}_2, \quad \frac{\partial \mathcal{L}}{\partial x_2} = -\frac{1}{2} k \frac{d(x_1 + L - x_2)^2}{d(x_1 + L - x_2)} \frac{\partial (x_1 + L - x_2)}{\partial x_2} = -k(x_1 + L - x_2)(-1) = k(x_1 + L - x_2)$$

$$\frac{d}{dt}(m \dot{x}_2) - k(x_1 + L - x_2) = 0$$

$$\ddot{x}_2 = \frac{k}{m}(x_1 + L - x_2)$$

4c. $x = x_1 - x_2, \quad U = \frac{1}{2} k (L + x_1 - x_2)^2 = \frac{1}{2} k (L + x)^2$

$$x_{cm} = \frac{1}{2}(x_1 + x_2)$$

$$\begin{cases} x_1 + x_2 = 2x_{cm} \\ x_1 - x_2 = x \end{cases} \quad \begin{aligned} 2x_1 &= 2x_{cm} + x \\ x_1 &= x_{cm} + \frac{x}{2} \end{aligned}$$

$$\begin{aligned} 2x_2 &= 2x_{cm} - x \\ x_2 &= x_{cm} - \frac{x}{2} \end{aligned}$$

$$2x_2 = 2x_{cm} - x$$

$$\dot{x}_1 = \dot{x}_{cm} + \frac{\dot{x}}{2} \quad \dot{x}_2 = \dot{x}_{cm} - \frac{\dot{x}}{2}$$

$$T = \frac{1}{2} m (\dot{x}_1^2 + \dot{x}_2^2) = \frac{1}{2} m \left(\left(\dot{x}_{cm} + \frac{\dot{x}}{2} \right)^2 + \left(\dot{x}_{cm} - \frac{\dot{x}}{2} \right)^2 \right)$$

$$= \frac{1}{2} m \left(\dot{x}_{cm}^2 + \frac{\dot{x}^2}{4} + \dot{x}_{cm}\dot{x} + \dot{x}_{cm}^2 + \frac{\dot{x}^2}{4} - \dot{x}_{cm}\dot{x} \right)$$

$$= \frac{1}{2} m \left(2\dot{x}_{cm}^2 + \frac{\dot{x}^2}{2} \right) = m \left(\dot{x}_{cm}^2 + \frac{\dot{x}^2}{4} \right)$$

$$T = m \left(\dot{x}_{cm}^2 + \frac{\dot{x}^2}{4} \right) \quad U = \frac{1}{2} k (L+x)^2$$

$$4d. \mathcal{L}(x_{cm}, x, \dot{x}_{cm}, \dot{x}) = m \left(\dot{x}_{cm}^2 + \frac{\dot{x}^2}{4} \right) - \frac{1}{2} k (L+x)^2$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}_{cm}} = 2m\dot{x}_{cm} \quad \frac{\partial \mathcal{L}}{\partial x_{cm}} = 0 \quad \frac{d}{dt}(2m\dot{x}_{cm}) - 0 = 0$$

$$\frac{d}{dt}(m\dot{x}_{cm}) = 0$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{m}{4}(2\dot{x}) = \frac{m}{2}\dot{x} \quad \frac{\partial \mathcal{L}}{\partial x} = -\frac{1}{2}k \frac{d(L+x)^2}{d(L+x)} \frac{d(L+x)}{dx}$$

$$\frac{d}{dt} \left(\frac{m}{2}\dot{x} \right) - (-k(L+x)) = 0 \quad = -k(L+x)$$

$$\frac{m}{2}\ddot{x} = -k(L+x) \quad \left[\ddot{x} = -\frac{2k}{m}(L+x) \right]$$

$$4e. \ddot{x} + \frac{2k}{m}x = -\frac{2k}{m}L \quad \text{Let } x(0) = x_0, \dot{x}(0) = v_0$$

where $\frac{k}{m} > 0$

$$x \propto e^{rt}$$

$$r^2 e^{rt} + \frac{2k}{m} e^{rt} = 0 \quad r^2 = -\frac{2k}{m} \quad r = \pm i \sqrt{\frac{2k}{m}}$$

$$x(t) = C_1 e^{i \sqrt{\frac{2k}{m}} t} + C_2 e^{-i \sqrt{\frac{2k}{m}} t}$$

$$= C_1 \left(\cos\left(\sqrt{\frac{2k}{m}} t\right) + i \sin\left(\sqrt{\frac{2k}{m}} t\right) \right)$$

$$+ C_2 \left(\cos\left(\sqrt{\frac{2k}{m}} t\right) - i \sin\left(\sqrt{\frac{2k}{m}} t\right) \right)$$

$$A = C_1 + C_2$$

$$B = i(C_1 - C_2)$$

$$x(t) = A \cos\left(\sqrt{\frac{2k}{m}} t\right) + B \sin\left(\sqrt{\frac{2k}{m}} t\right) \quad \text{Homogeneous solution}$$

$$x_p = Ct + D \quad \ddot{x}_p = 0$$

$$0 + \frac{2k}{m}(Ct + D) = -\frac{2kL}{m} \quad C = 0 \quad \text{particular solution}$$

$$\frac{2kD}{m} = -\frac{2kL}{m} \quad D = -L \quad x_p = -L$$

$$x(t) = A \cos\left(\sqrt{\frac{2k}{m}} t\right) + B \sin\left(\sqrt{\frac{2k}{m}} t\right) - L$$

$$x_0 = A - L \quad A = x_0 + L$$

$$x(t) = -A \sqrt{\frac{2k}{m}} \sin\left(\sqrt{\frac{2k}{m}} t\right) + B \sqrt{\frac{2k}{m}} \cos\left(\sqrt{\frac{2k}{m}} t\right)$$

$$v_0 = \dot{x}(0) = B \sqrt{\frac{2k}{m}} \quad B = v_0 \sqrt{\frac{m}{2k}}$$

$$x(t) = (x_0 + L) \cos\left(\sqrt{\frac{2k}{m}} t\right) + v_0 \sqrt{\frac{m}{2k}} \sin\left(\sqrt{\frac{2k}{m}} t\right) - L$$

There is oscillatory motion similar to SHO but the displacement is shifted downwards by distance L . Sinusoidal motion for $x(t)$.

$\frac{dx_{cm}}{dt} = 0$ so $\dot{x}_{cm}$ is a constant thus

x_{cm} can be a linear function

$$x_{cm} = \alpha t + \beta = \dot{x}_{cm}(0) t + x_{cm}(0)$$

$$\dot{x}_{cm} = \frac{dx_{cm}}{dt}$$